



Topology

PC2 sent a ping to PC4.

- PC2 will send an ICMP Echo Request with the source IP address of 192.168.1.2 and destination IP address of PC4 which is 192.168.1.4. The source MAC address is its own, but the destination MAC address associated with PC4 is unknown, hence the ICMP Echo Request will be buffered and address resolution protocol will be performed first.
- PC2 will start with an ARP request asking for the MAC address of 192.168.1.4, this also being the destination MAC address of the request. SW1 will be the first to receive this packet – it will list its destination MAC address and associate it with its F0/1 interface in its MAC address table. SW1 will proceed to flood the ARP request out of its F0/2 and G0/1 interfaces.
- PC2 and SW2 will be the next to receive the ARP request. PC2 will ignore it, whereas SW2 will then note down the source MAC address and associate it with its G0/1 interface in its own MAC address table. Afterwards, it will be flooded out of its F0/1 and F0/2 interfaces to PC3 and PC4 respectively.
- PC3 ignores the packet, but PC4 accepts it and prepares an ARP reply, with the source MAC address as its own MAC address and the destination MAC address as PC1's MAC address. This ARP reply will first reach SW2, which will know to reach out to PC1 through its G0/1 interface thanks to its MAC address table.
- SW1 will be the next hop node, and it will also know to send it out of its F0/1 interface thanks to its MAC address table. The ARP reply finally reaches PC1, and PC1 reads its source MAC address, and lists that down in its ARP table corresponding to 192.168.1.4.
- The recently sourced MAC address will be appended as the destination MAC address of the ICMP Echo Request, which will then be forwarded only to SW1, then SW2, and PC4.
- PC4 reads the message and sends an ICMP Echo Reply to SW2, SW1, then back to PC1.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=12ms TTL=128
Reply from 192.168.1.4: bytes=32 time=6ms TTL=128
Reply from 192.168.1.4: bytes=32 time=6ms TTL=128
Reply from 192.168.1.4: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 12ms, Average = 7ms

```

Ping 192.168.1.4 - PC2 pings PC4

```

SW2>
SW2>en
SW2>enable
SW2#show mac add
SW2#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
1       0001.647b.3119   DYNAMIC   Fa0/2
1       0004.9a6e.d870   DYNAMIC   Fa0/1
1       0060.5c56.14d3   DYNAMIC   Gig0/1
1       00d0.d3ad.9cab   DYNAMIC   Gig0/1
SW2#

```

MAC address table of SW1 after the ping

```

SW1>
SW1>en
SW1>en
SW1#show mac ad
SW1#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
1       0001.647b.3119   DYNAMIC   Gig0/1
1       0004.9a6e.d870   DYNAMIC   Gig0/1
1       0060.5c56.14d3   DYNAMIC   Fa0/2
1       00d0.d3ad.9cab   DYNAMIC   Fa0/1

```

MAC address of SW2 after the ping

```

SW2#clear mac address-table dynamic
SW2#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----

```

Clear mac address-table dynamic - Clearing the MAC addresses of the SW2

```
SW1#clear mac address-table dynamic
SW1#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
-----

```

Clear mac address-table dynamic - Clearing the MAC addresses of the SW1